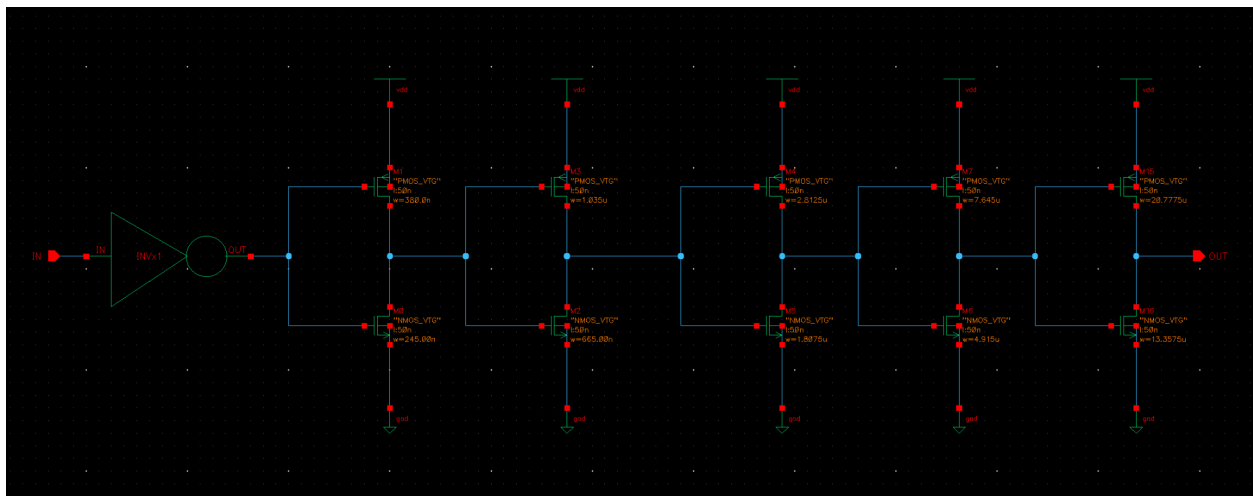


EECS 217 - VLSI System Design Project 2

Repeater Chain



My repeater chain consists of 6 stages. Given $CL=1.28\text{pF}$, I needed to find the C_{in} of my minimum-sized inverter.

Finding C_{in} :

To find C_{in} , I tried two different methods.

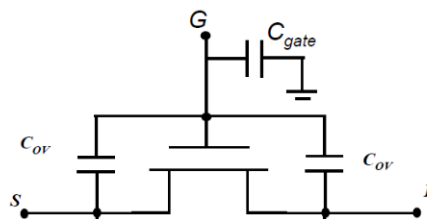
1. The first method was to use the C_{gate} equation below to add up the C_{gate} for both the PMOS and NMOS:

gate capacitance:

$$C_{gate} = \frac{\epsilon}{t_{ox}} WL$$

overlap capacitance:

$$C_{ov} = \frac{\epsilon}{t_{ox}} W X_{ov}$$



t_{ox} of the NMOS and PMOS was found in the design files of the PDK

```

PMOS_VTG.inc  NMOS_VTG.inc
1  * Customized PTM 45 NMOS nom
2
3  .model NMOS_VTG nmos level = 54
4
5  * parameters related to the technology node
6  +tnom = 27      epsrox = 3.9
7  +eta0 = 0.006   nfactor = 2.1   wint = 5e-09
8  +cgso = 1.1e-10 cgdo = 1.1e-10 xl = -2e-08
9
10 * parameters customized by the user
11 +tox = 1.14e-09  toxp = 1e-09   toxm = 1.14e-09  toxref = 1.14e-09
12 +dtoc = 0.14e-09 lint = 3.75e-09
13 +vth0 = 0.4106   k1 = 0.4    u0 = 0.045   vsat = 123000
14 +rdsw = 155      ndep = 3.4e+18 xj = 1.98e-08
15

```

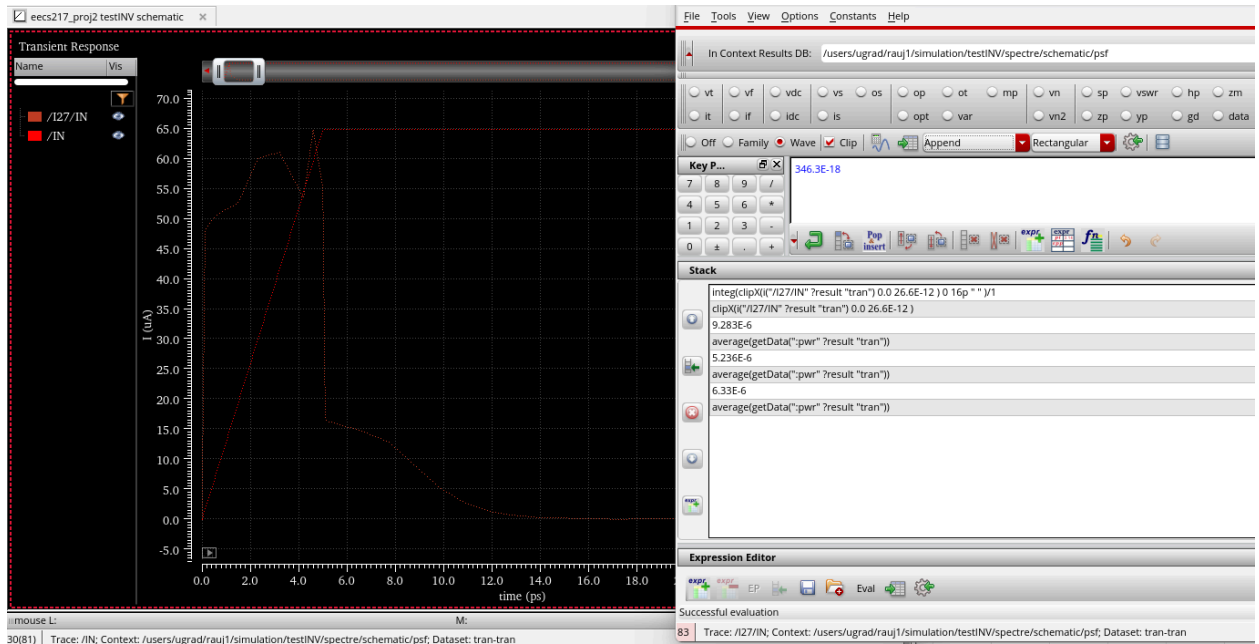
ξ was treated as a given since

$$\xi = \xi_o * \xi_r = 3.9 * 8.854 * 10^{-12} = 3.45 * 10^{-11}$$

Therefore, my Cgate_nmos=1.92E-16 and Cgate_pmos=1.36E-16. Adding these two values up only resulted in a Cin=0.328fF, much lower than expected.

- The second method was to measure the Cin using $C = \frac{1}{V_c} \int_0^t ic(t)dt$, where Vc=1V and

$\int_0^t ic(t)dt$ is found by measuring the input current during Vin=0→1



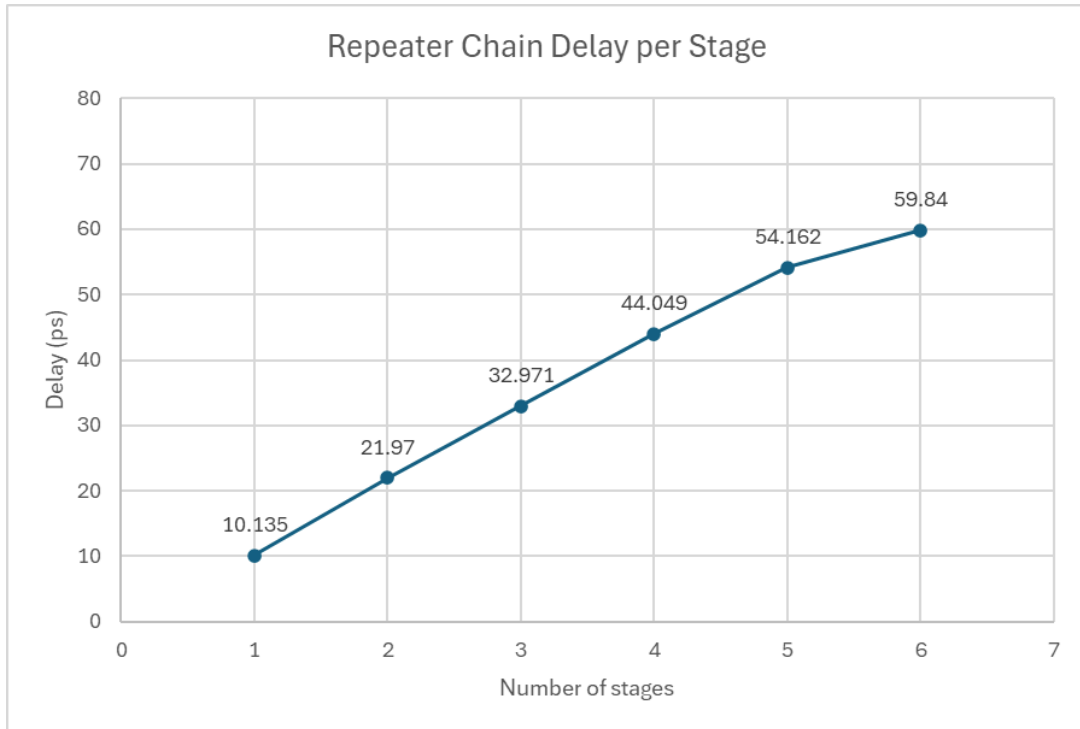
Again, I received only Cin=0.3463fF, below the expected value.

Since I could not determine the Cin, I approximated it to be 3fF. Therefore:

$$N = \ln(1.28pF/3fF) = \sim 6 \text{ stages}$$

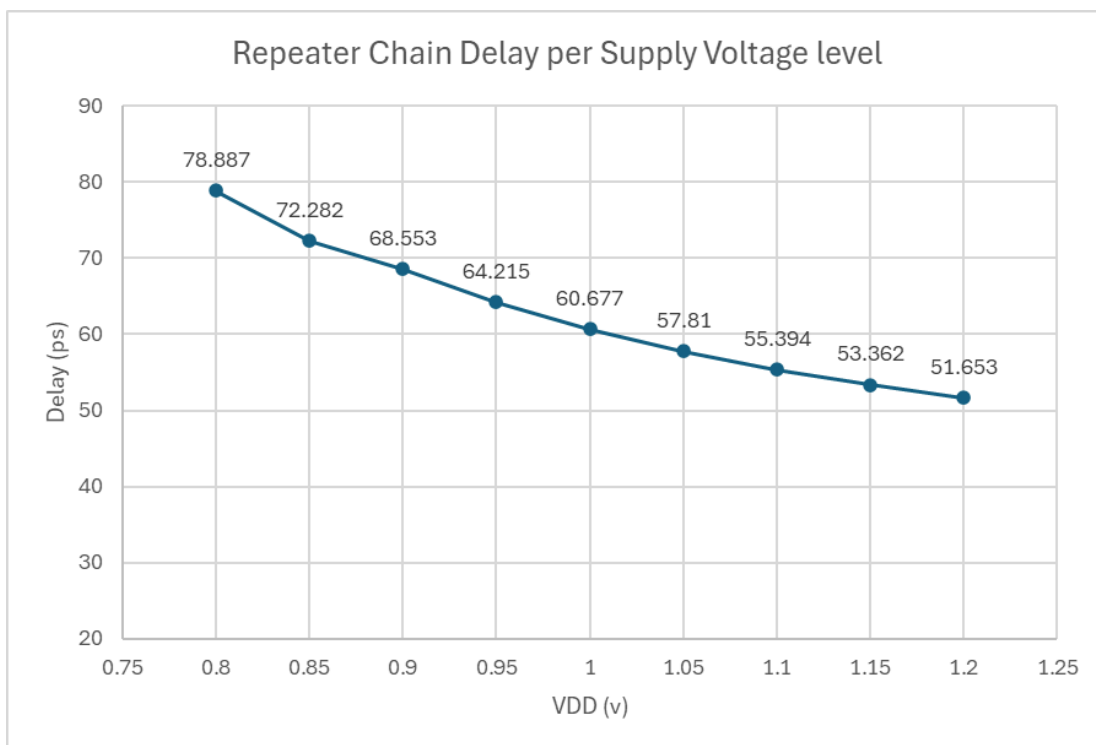
$$u = (1.28pF/3fF)^{1/6} = \sim 3 \text{ for scaling}$$

The width of my last NMOS and PMOS stage: $W_{NMOS}=13.357\mu m$, $W_{PMOS}=20.777\mu m$



1.

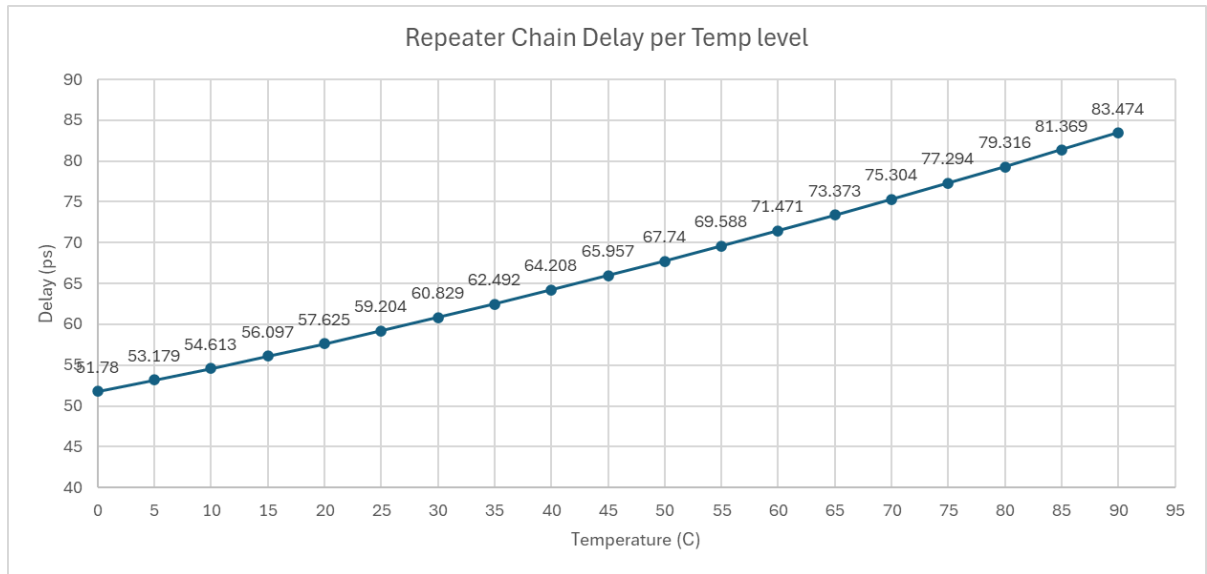
As the number of stages increases, the delay also increases, as expected



2.

As the voltage level increases, the delay decreases, as expected

3.



As the temperature level increases, the delay increases, as expected

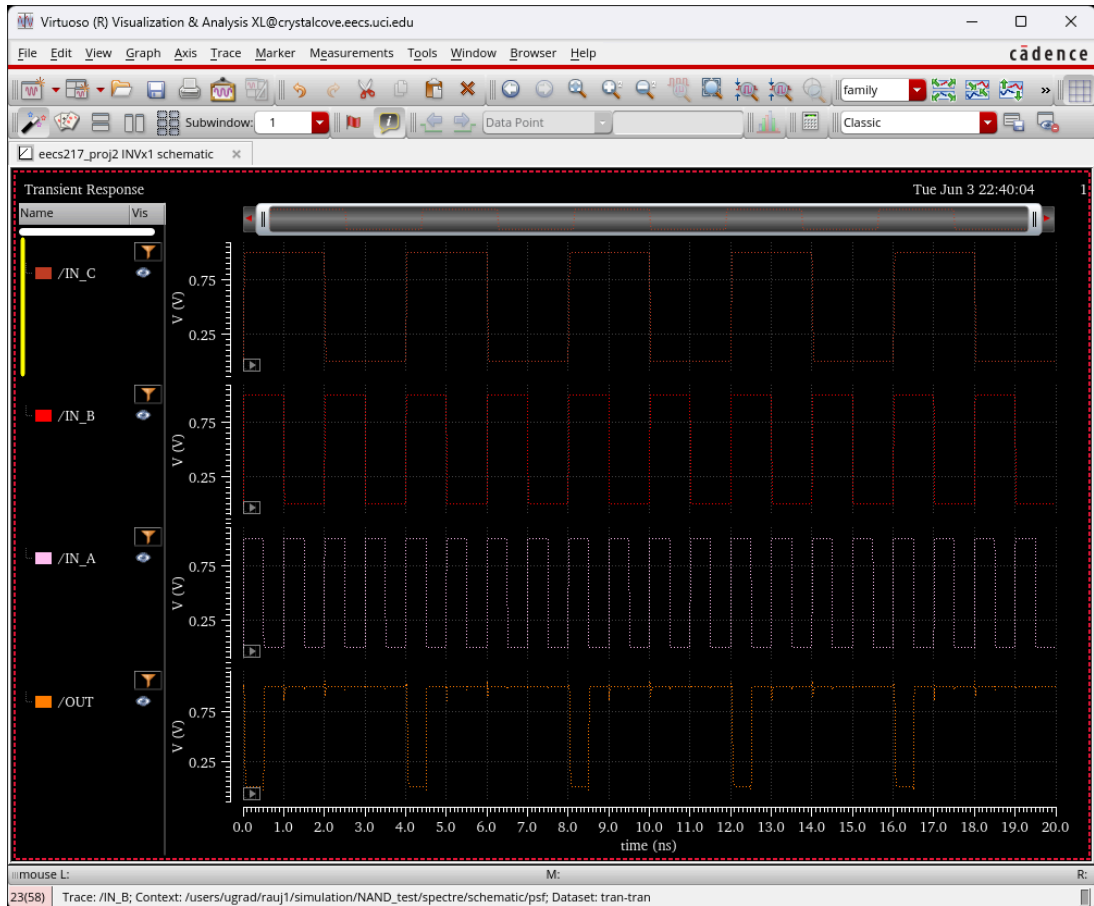
Cell Library

- The $W_{\text{Min_PMOS}}$ was found to be 140nm for $t_{\text{rise}} \approx t_{\text{fall}}$
- NOTE: Power dissipation for all circuits is found by injecting the 1V input at 1GHz and taking the average power being plotted

1.

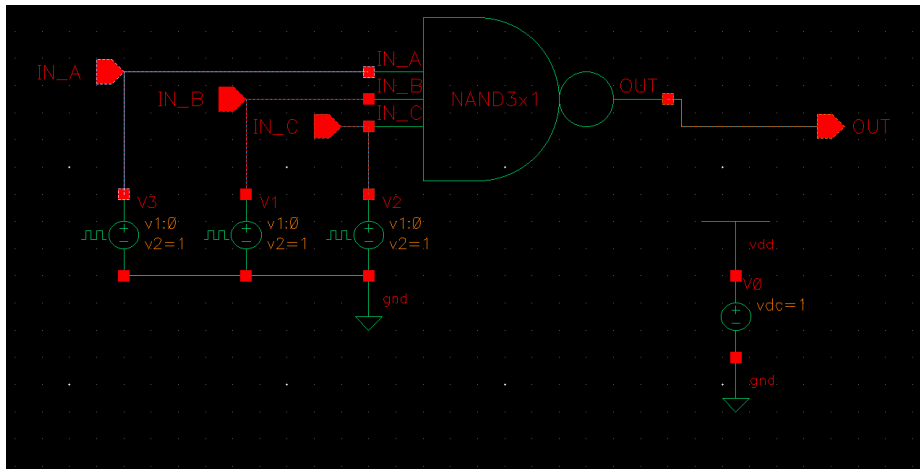
Logic Gates:	tr (ps)	tf (ps)	td (ps)	Pd (uW)
INVx1	6.094	5.795	5.22	0.5287
INVx2	5.079	4.667	4.339	0.8728
NAND2x1	5.203	14	4.419	0.8837
NAND2x2	4.826	7.992	4	1.506
NOR2x1	7.565	4.939	6.378	0.7515
NOR2x2	6.011	4.23	5.467	1.223
NAND3x1	5.38	22.974	4.169	1.226
NAND3x2	4.271	21.065	3.627	2.133
NOR3x1	27.761	4.354	16.754	1.162
NOR3x2	20.373	3.574	14.608	2.042
XOR2x1	11.984	16.063	36.944	4.131

XOR2x2	9.47	12.306	30.985	7.103
XNOR2x1	9.702	7.746	45.643	4.951
XNOR2x2	7.728	6.028	44.116	8.595

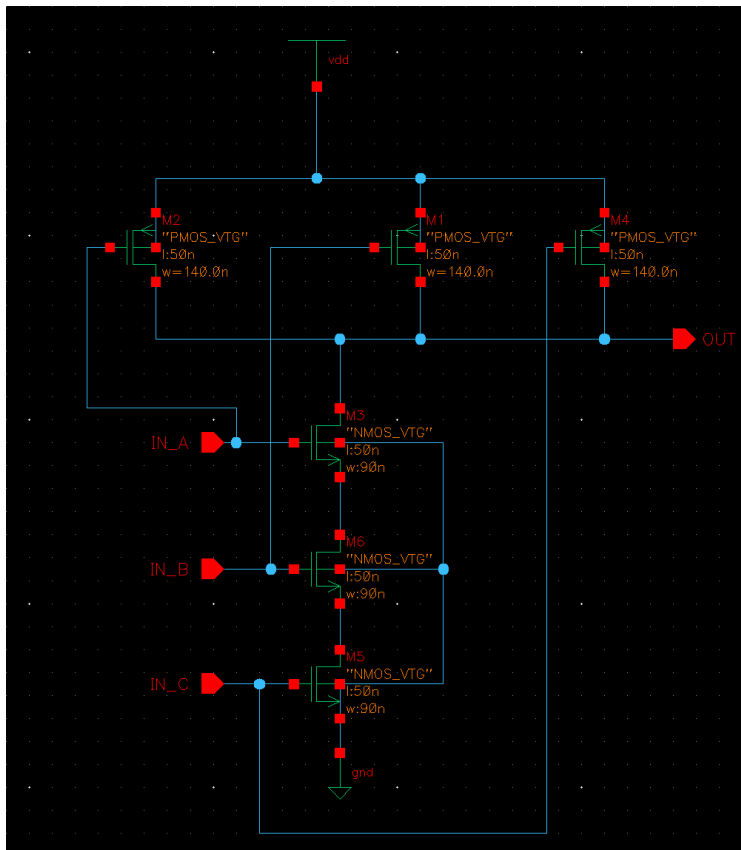


2.

The screenshot above shows the correct functionality of the NAND3x1 gate



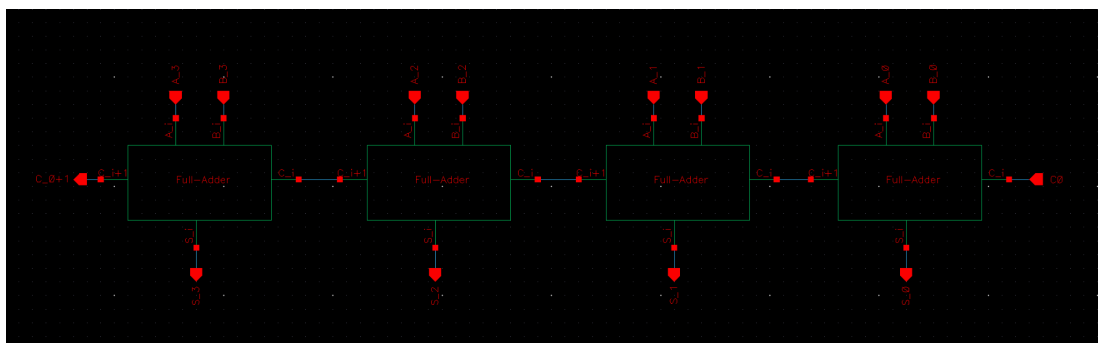
Testbench of the functionality test



Schematic-level view of the NAND3x1

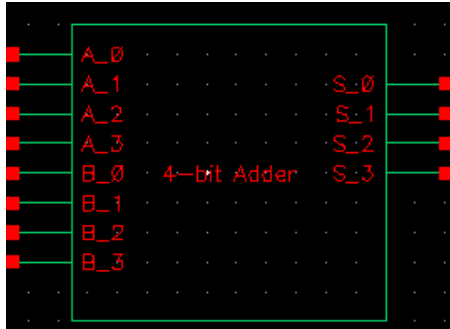
4-bit Adder

For my 4-bit adder, I selected the 4-bit ripple carry adder from lecture 15, as I was already familiar with it, and its delay increases only linearly with the number of bits.

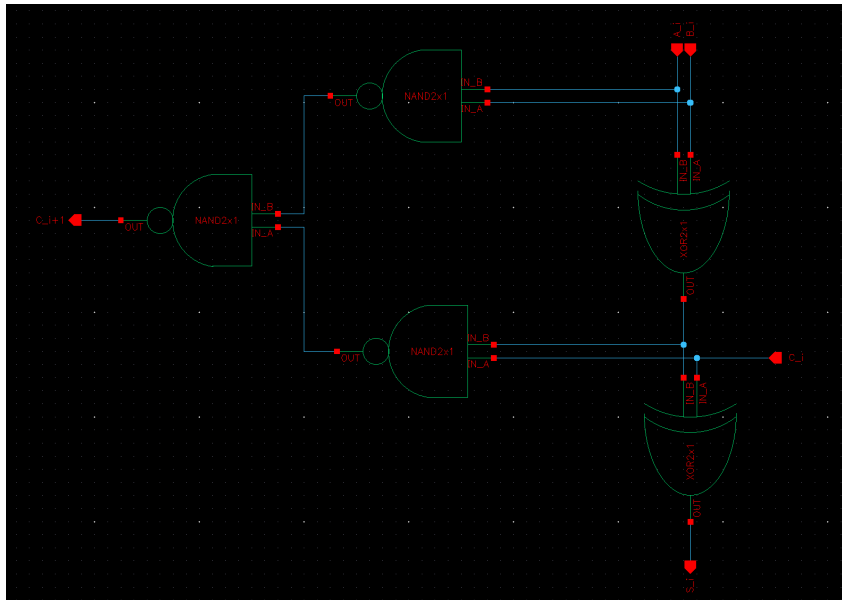


1.

Block-level diagram of the 4-bit adder. The Cin of the first adder and Cout of the last adder were later removed

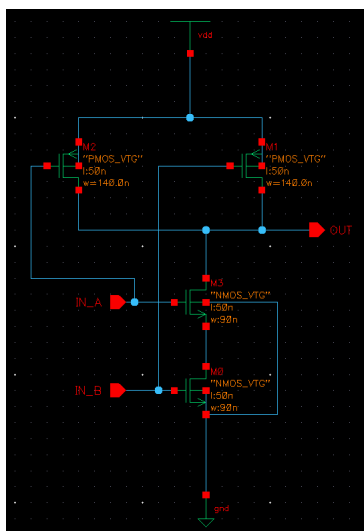


4-bit Adder symbol

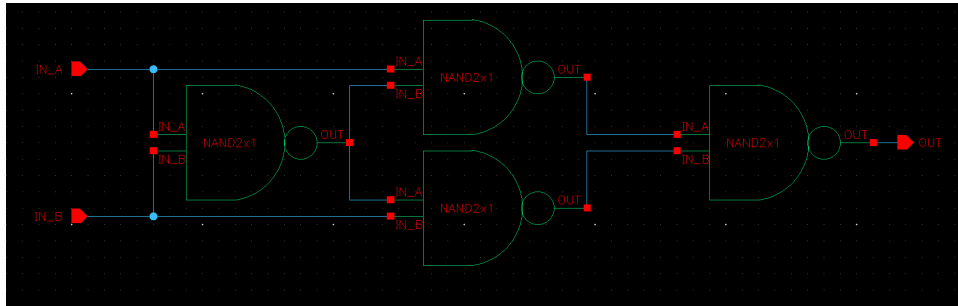


2.

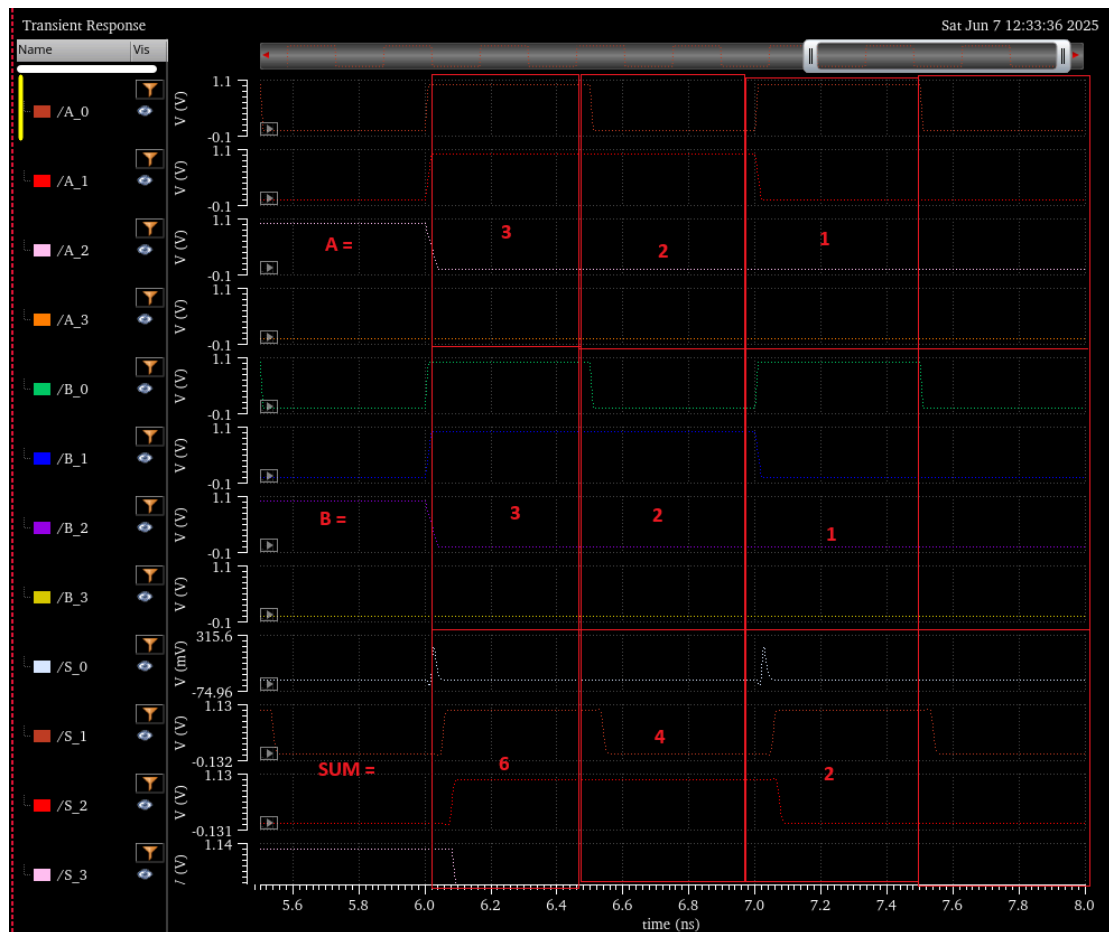
1-bit Full Adder circuit:



NAND2x1 circuit used



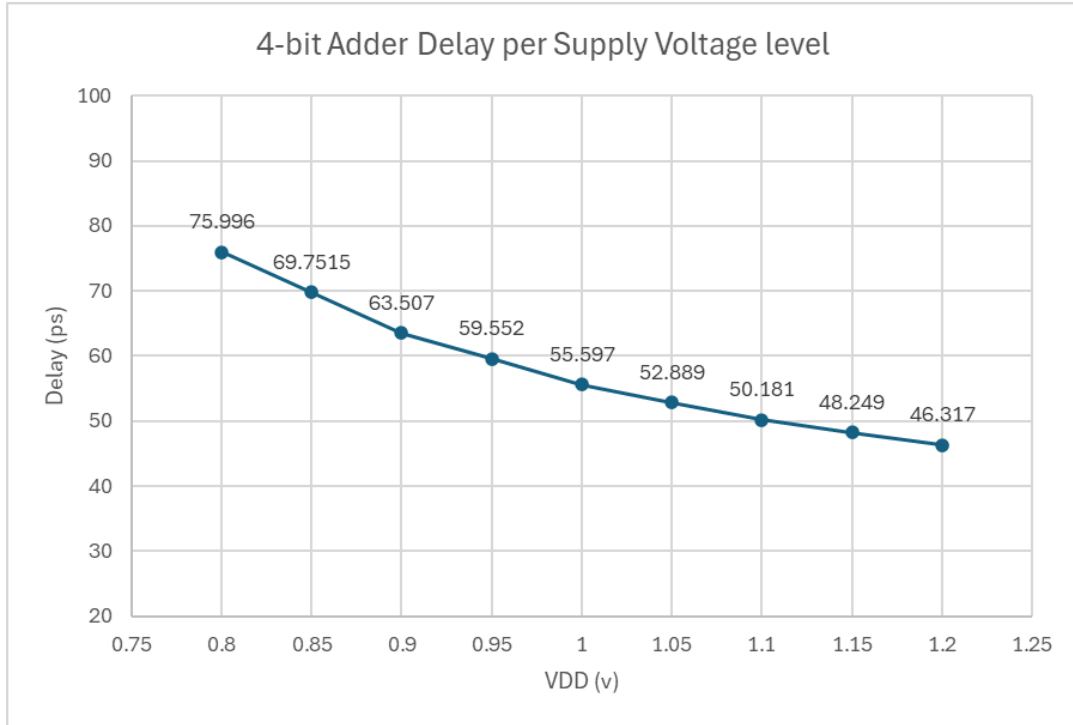
XOR2x1 circuit used



3.

Functionality check of the 4-bit adder on 3 examples

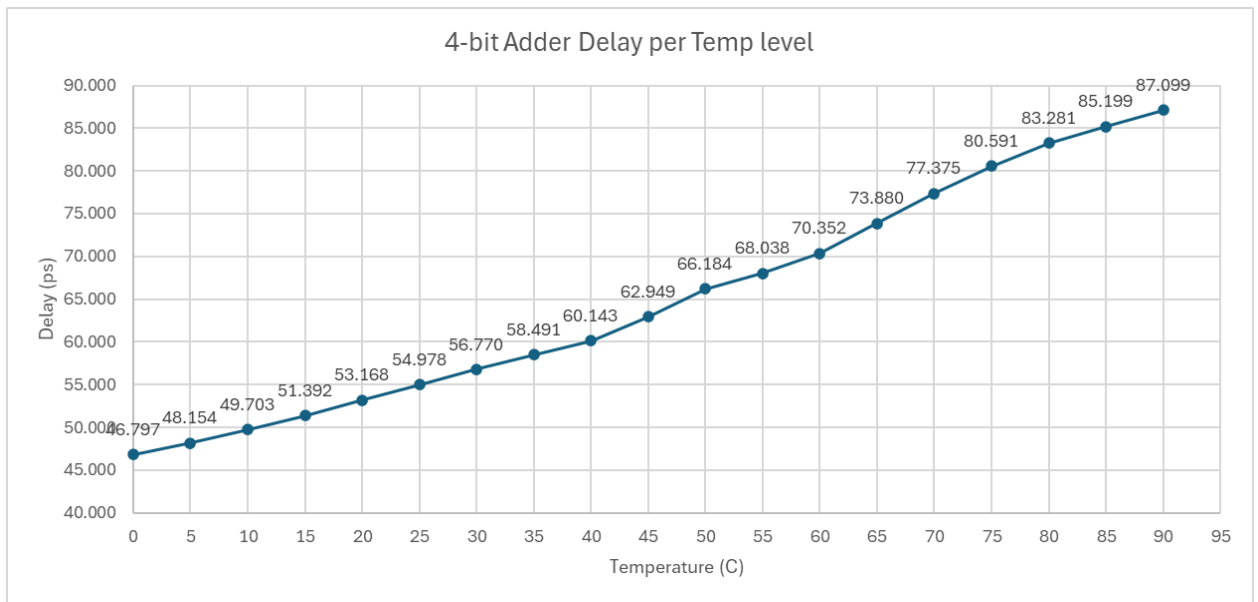
4. The critical path for the adder is from the input A_0 to the output S_3. The delay is 55 picoseconds. A=1111, B=1111 causes the worst-case delay
5. Power Dissipation = 14.2uW
 - NOTE: Power dissipation for all circuits is found by injecting the 1V input at 1GHz and taking the average power being plotted



6.

As the voltage level increases, the delay decreases, as expected

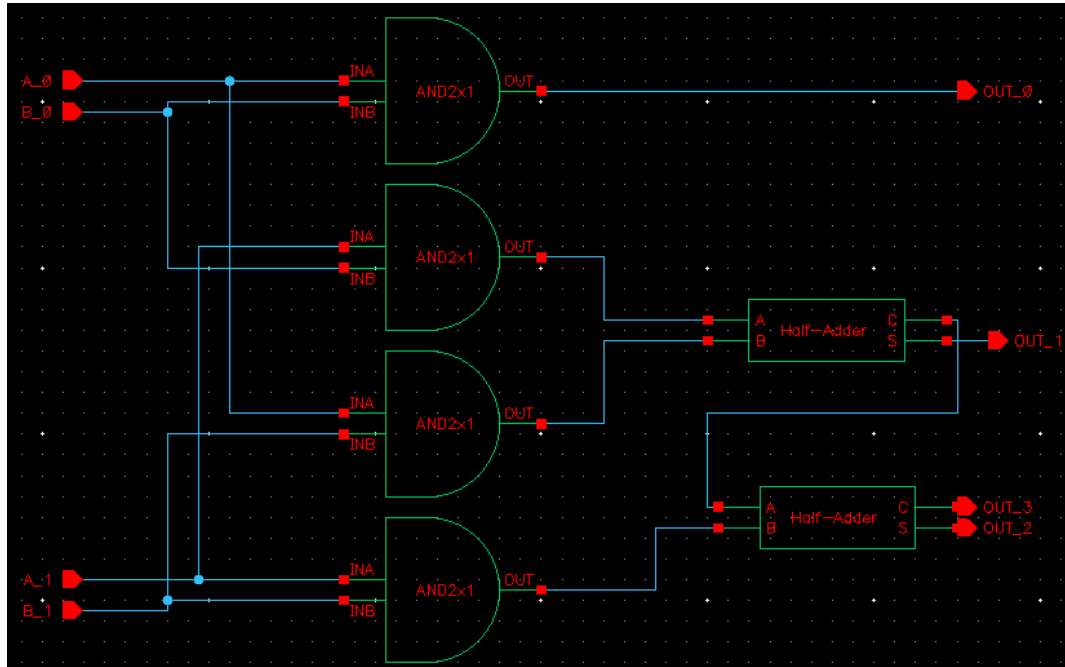
7.



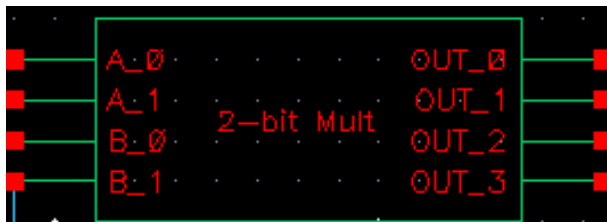
As the temperature level increases, the delay increases, as expected

2-bit Multiplier

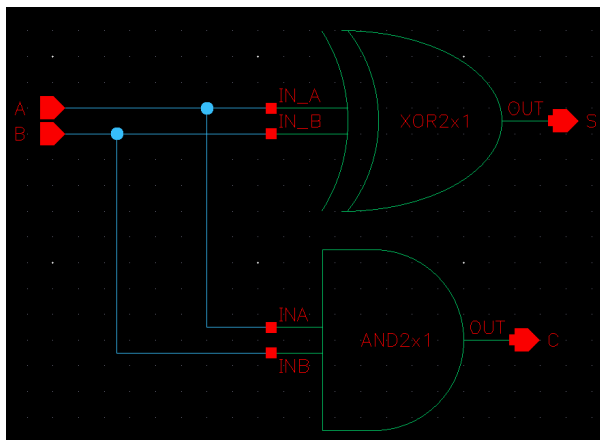
My 2-bit multiplier uses a combination of AND gates and Half-adders. I chose this architecture since the critical path for worst-case delay is only going through 2 AND gates and 1 XOR gate.



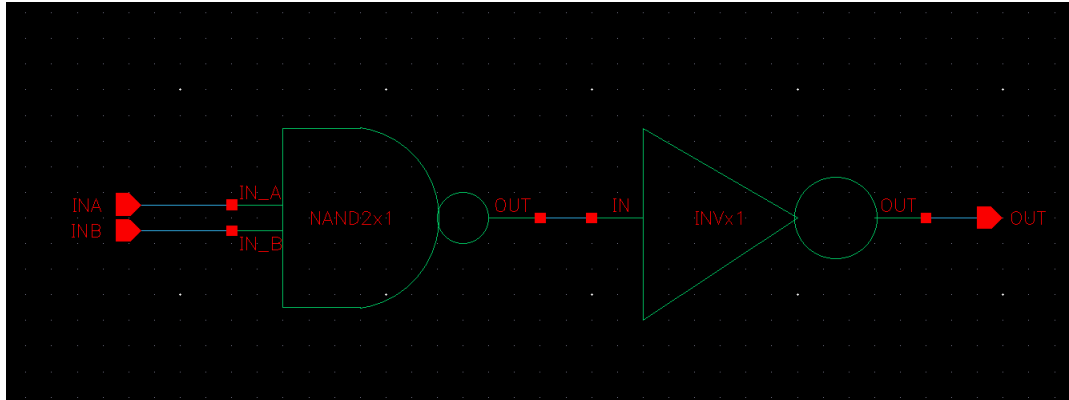
1. Block-level diagram of the 2-bit multiplier



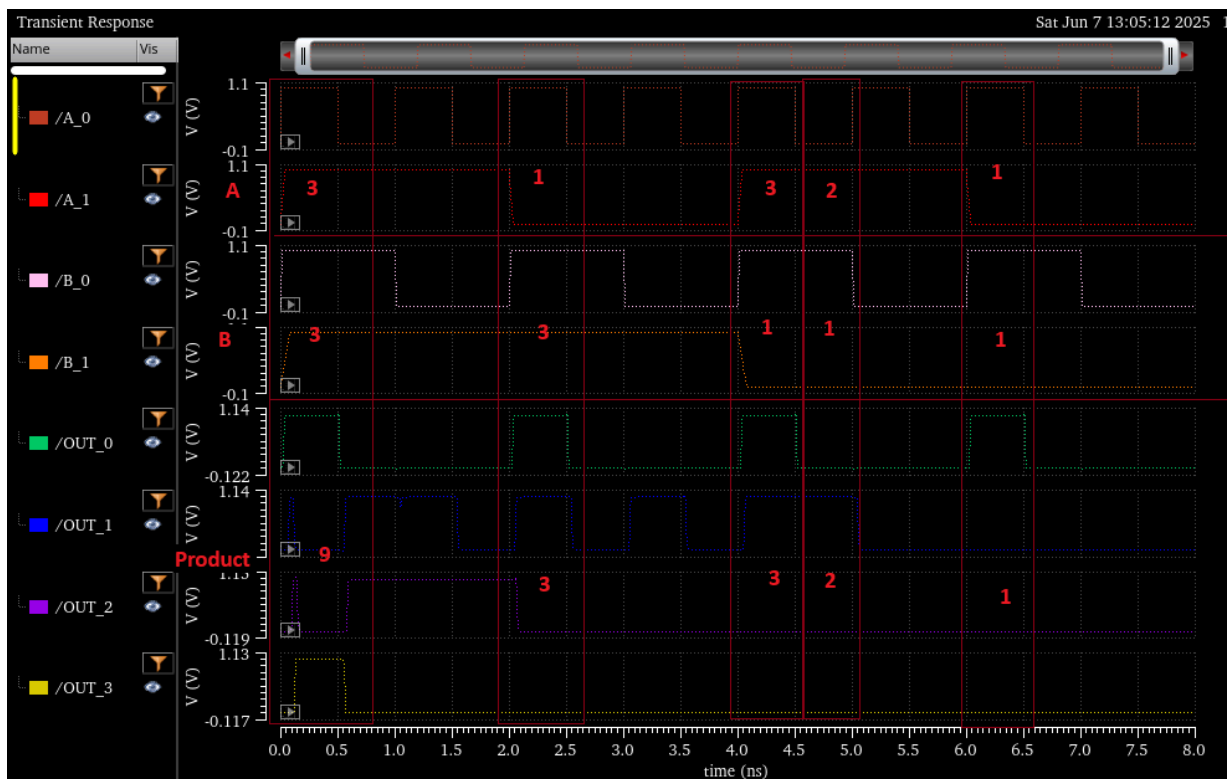
2-bit multiplier symbol



Half-adder circuit used in the 2-bit multiplier

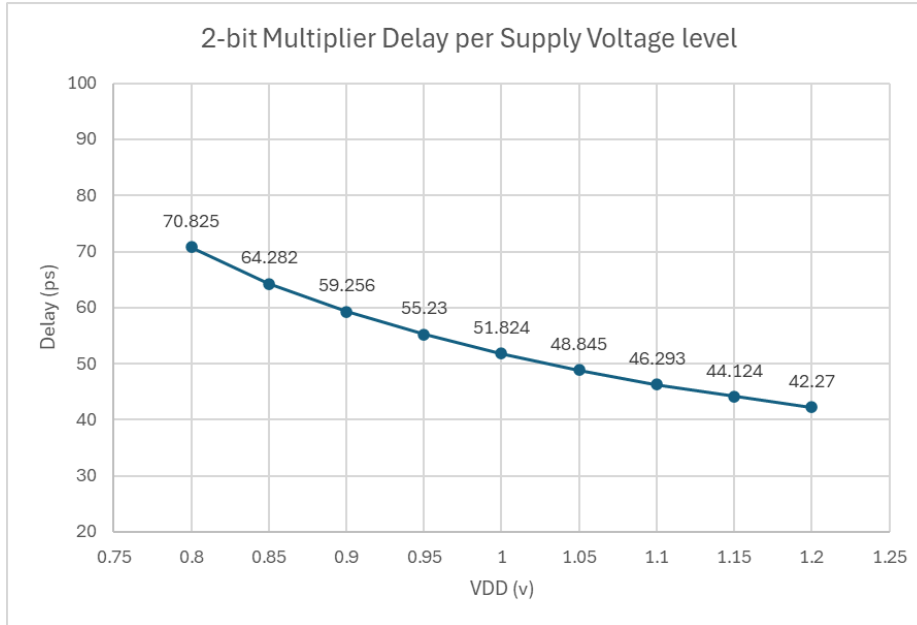


2. Single-bit multiplier (AND gate)



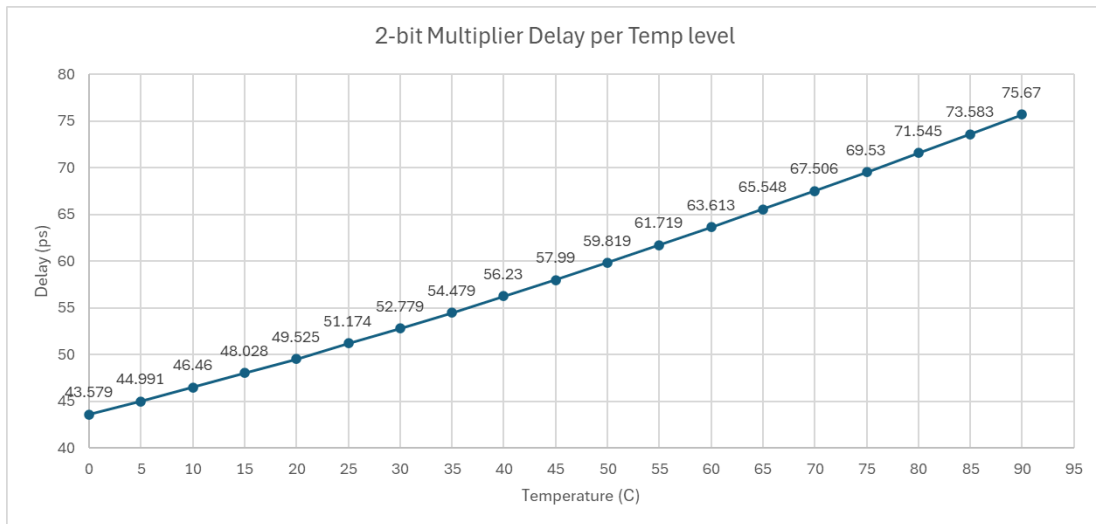
3. Functionality check of 2-bit multiplier using 4 multiplication examples

4. The critical path for the adder is from the input A_0 to the output OUT_3, as it goes through 2 AND gates and 1 XOR gate. The delay is 51.81 picoseconds. A=11, B=11 causes the worst-case delay
5. Power Dissipation = 6.86uW
 - NOTE: Power dissipation for all circuits is found by injecting the 1V input at 1GHz and taking the average power being plotted



6.

As the voltage level increases, the delay decreases, as expected

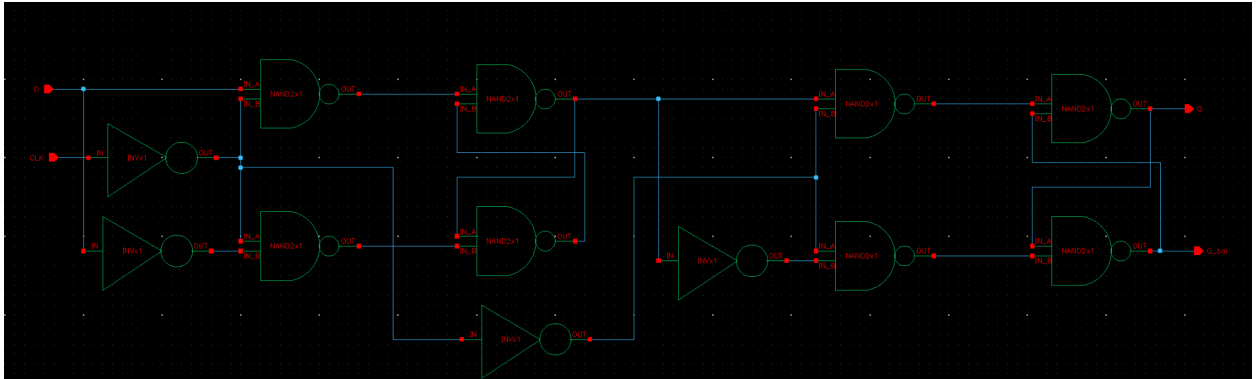


7.

As the temperature level increases, the delay increases, as expected

2-bit and 4-bit Registers

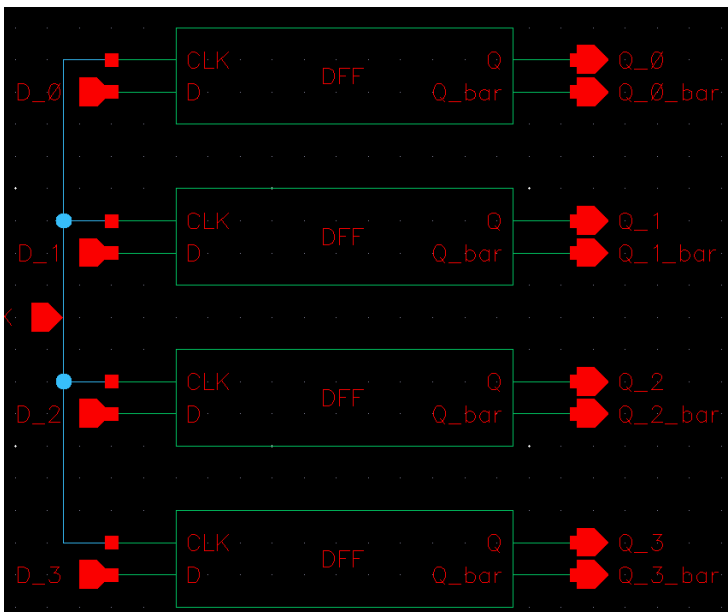
1.



FF circuit schematic. My design is a positive edge-triggered DFF.



2-bit Register



4-bit Register

2. setup time of the FF:

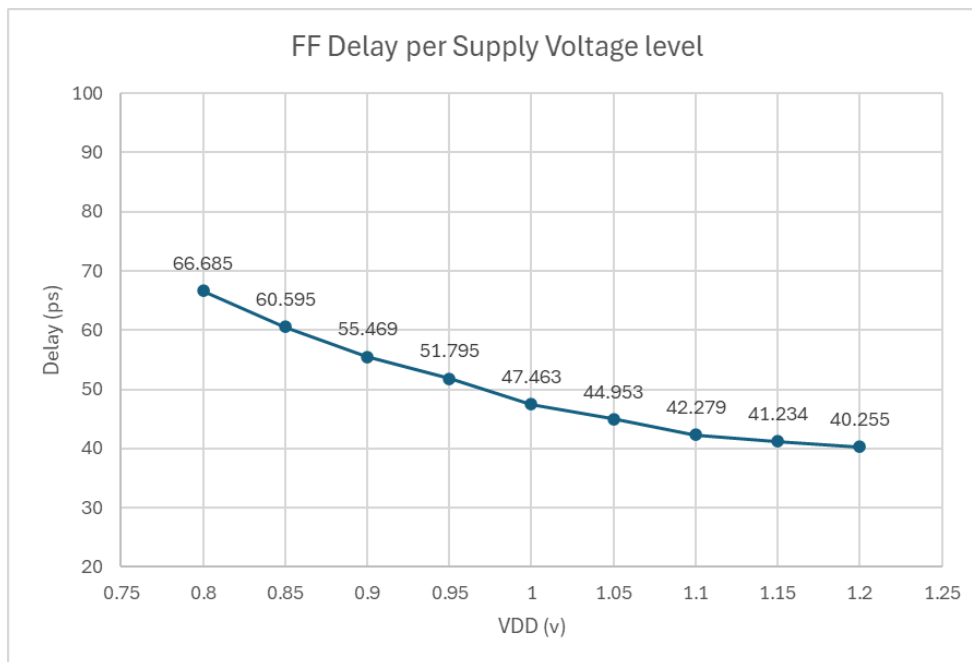
setup time:	18.4ps
period:	1ns
15% of the period:	150ps

$$18.4\text{ps} < 150\text{ps}$$

3. clock-to-Q delay of the FF:

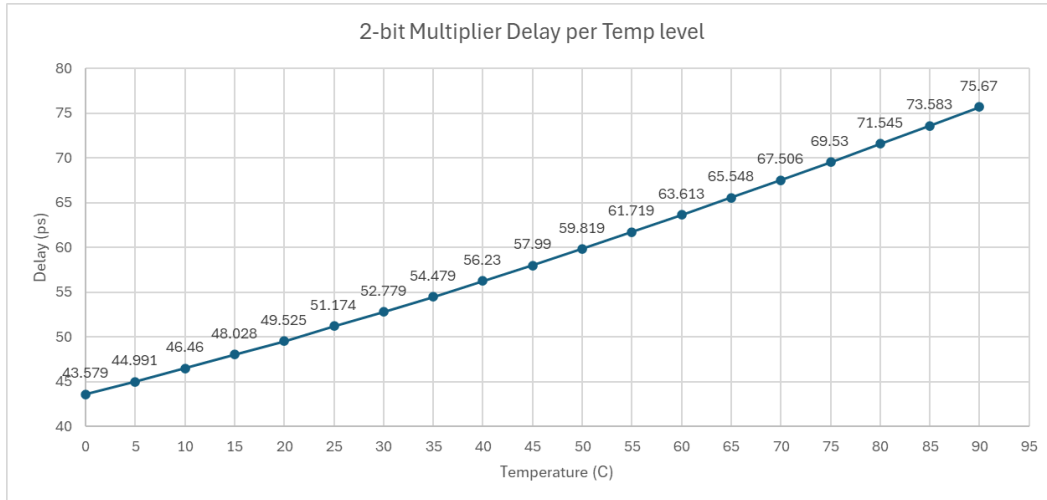
clk to Q:	48.276ps
period:	1ns
15% of the period:	150ps

$$48.276\text{ps} < 150\text{ps}$$



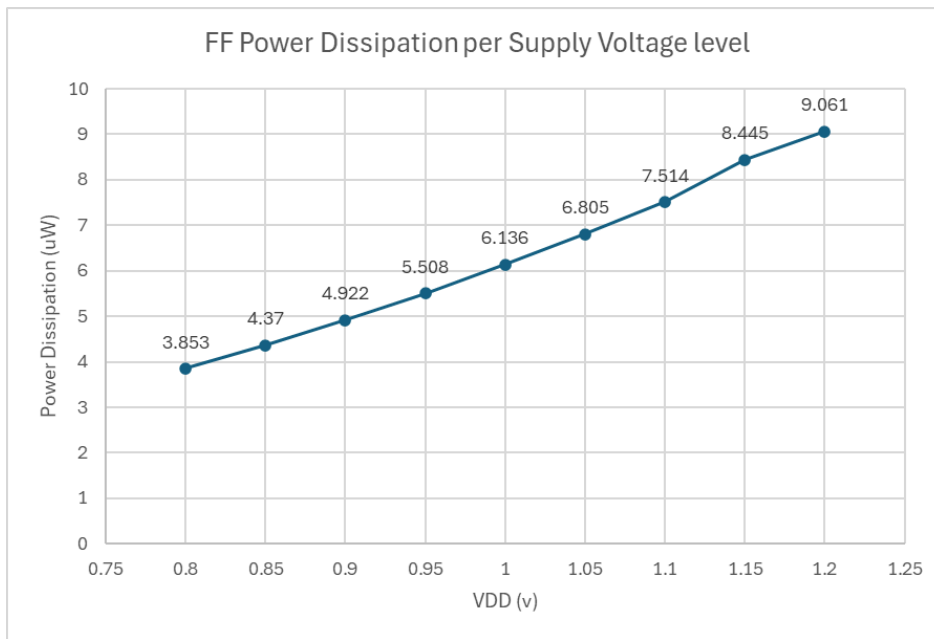
- 4.

As the voltage level increases, the delay decreases, as expected



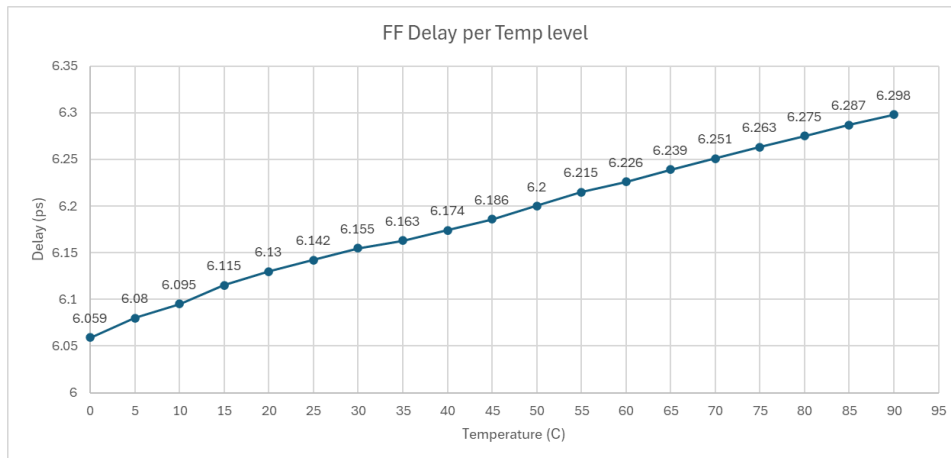
5.

As the temperature level increases, the delay increases, as expected



6.

Applied a pulse train at the input of the FF (0→1→0 etc.), where the value of the input changes once every clock cycle. Plotted the power dissipation of the FF vs. supply voltage.

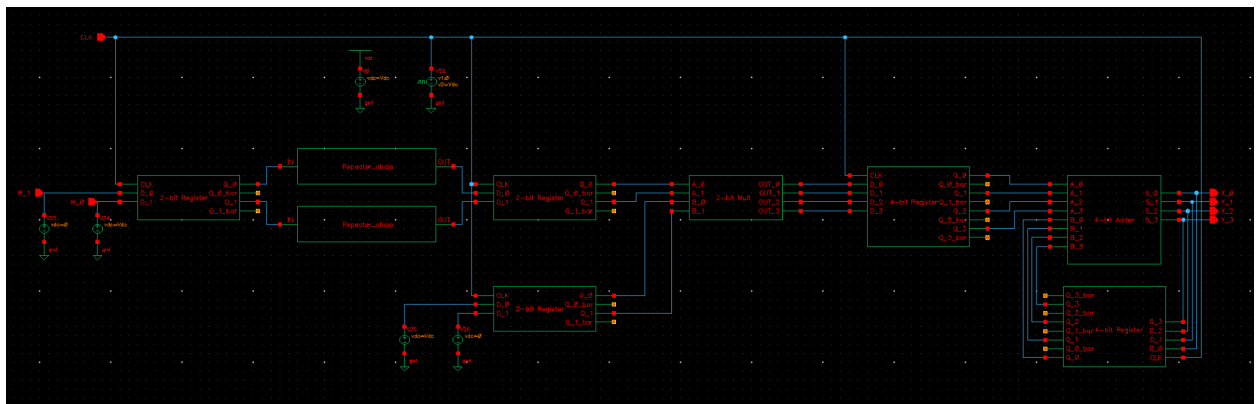


7.

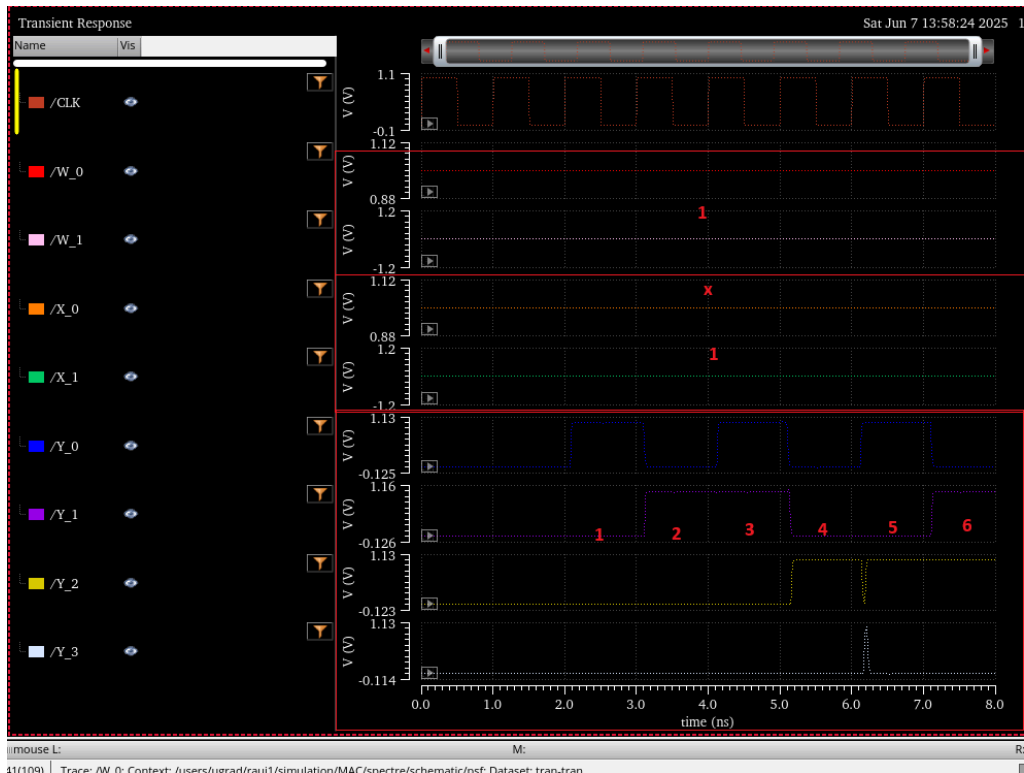
For the same pulse train input as in the previous item, the power dissipation of the FF vs. temperature is plotted.

System Integration

1.

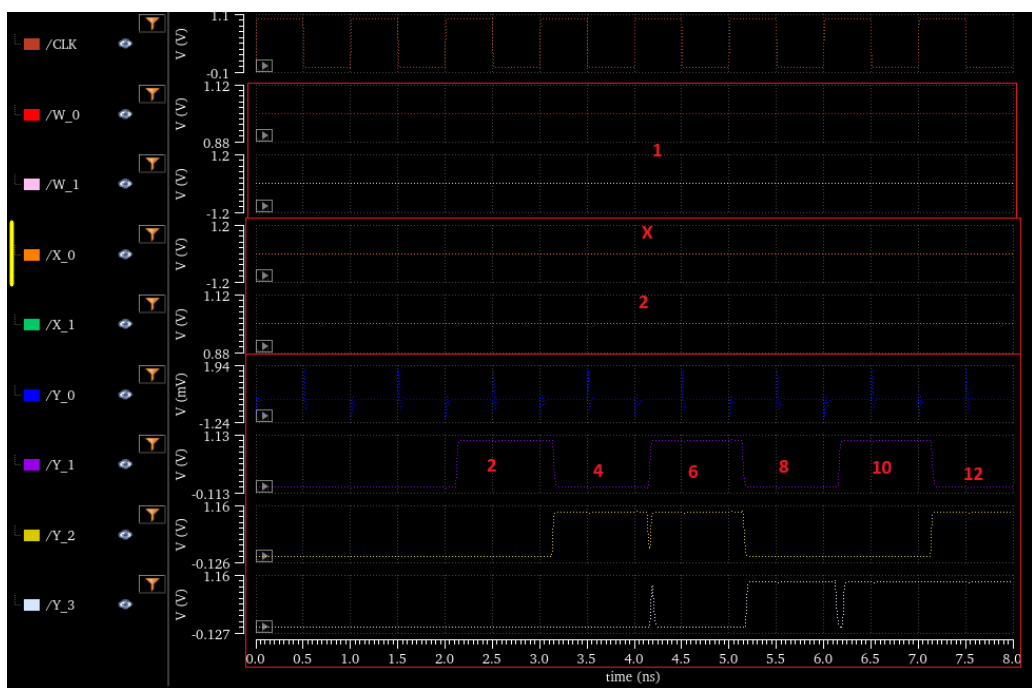


Block-level schematic of the MAC design

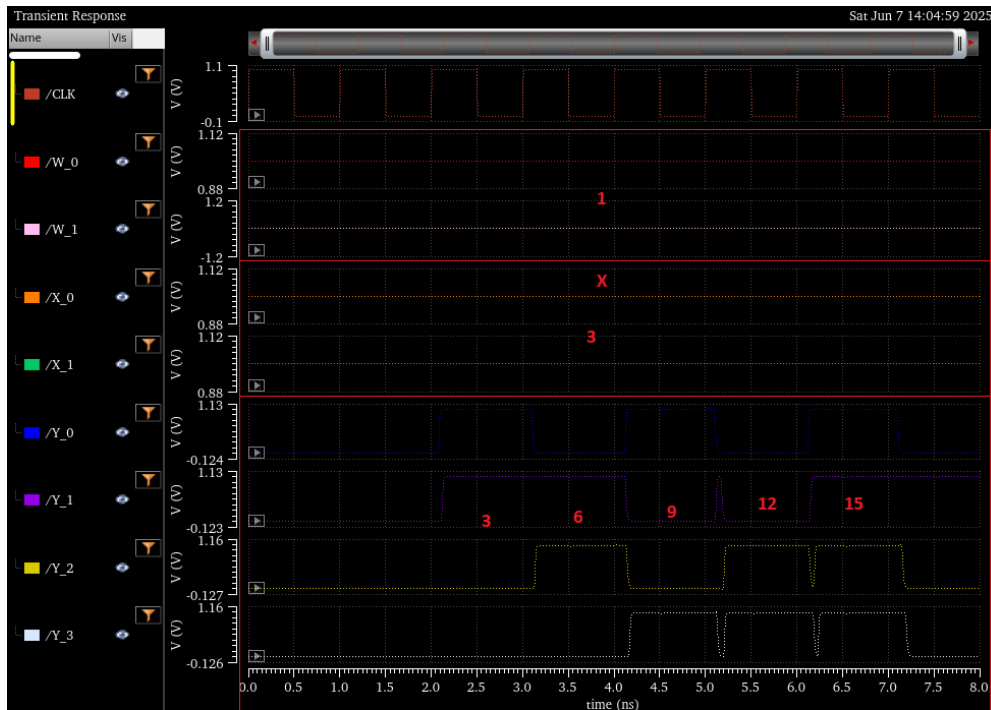


2.

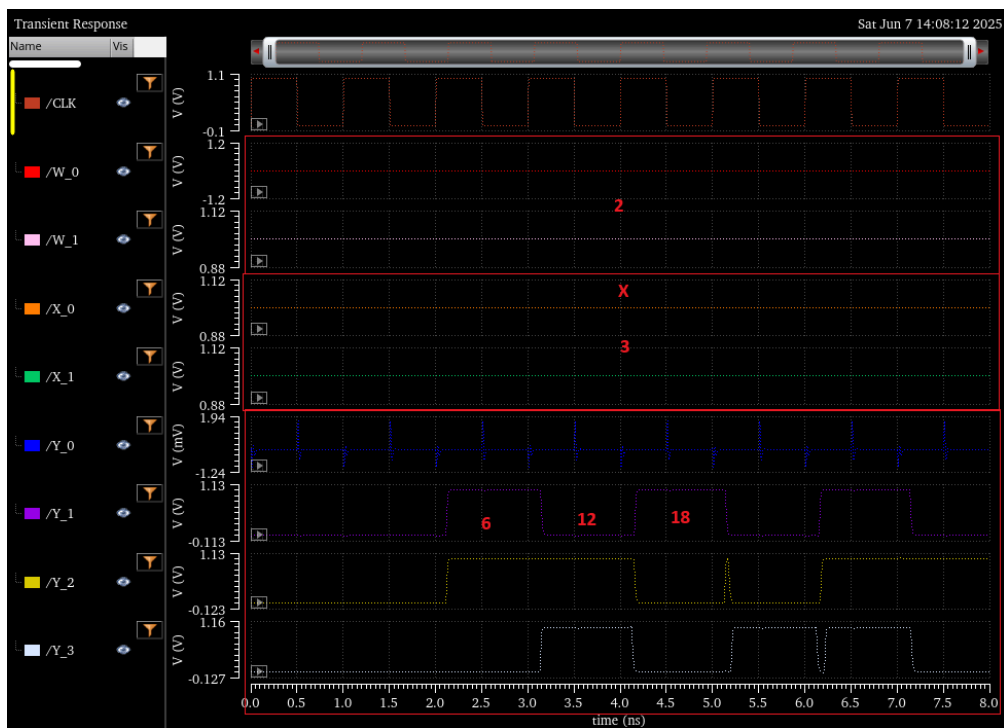
1x1 example:



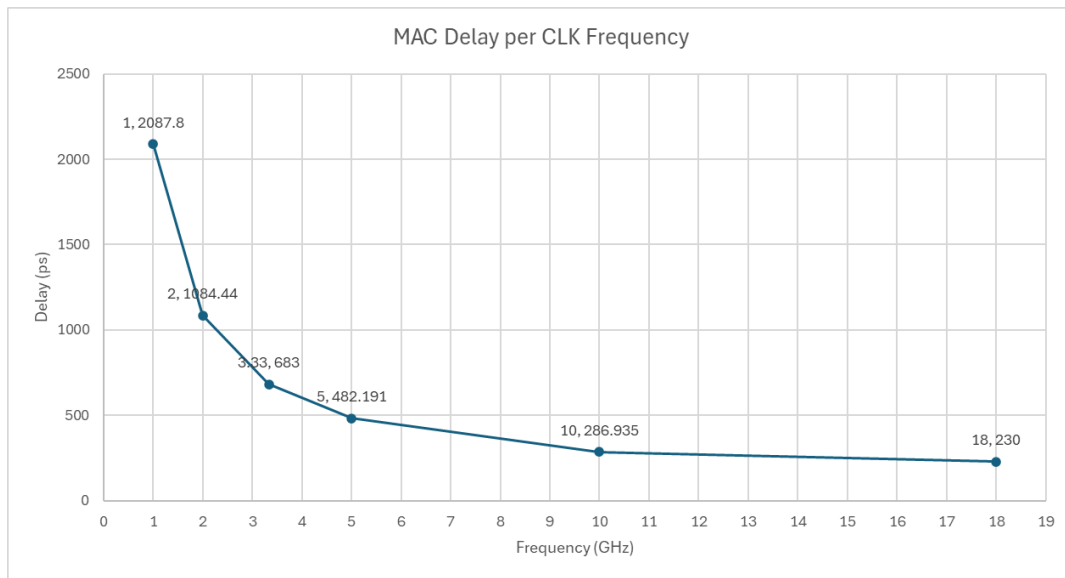
2x1 example:



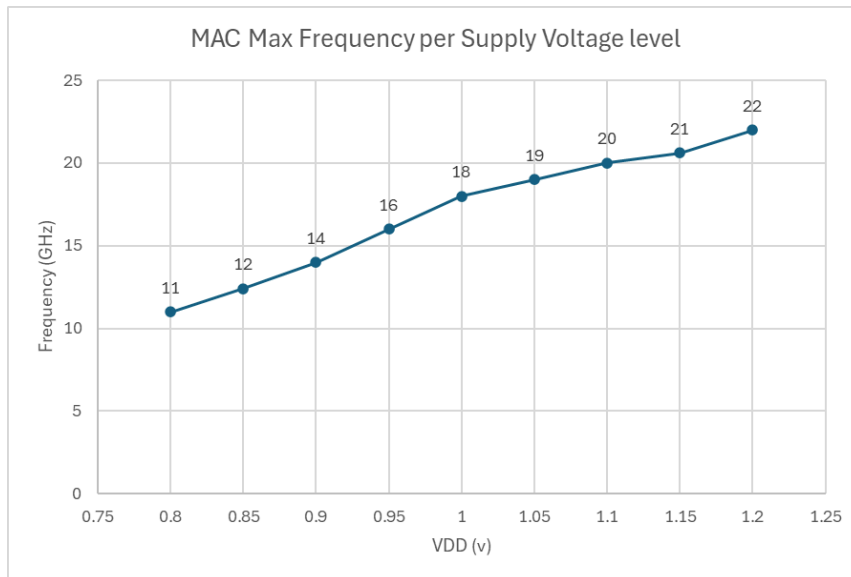
3x1 example:



3x2 example:

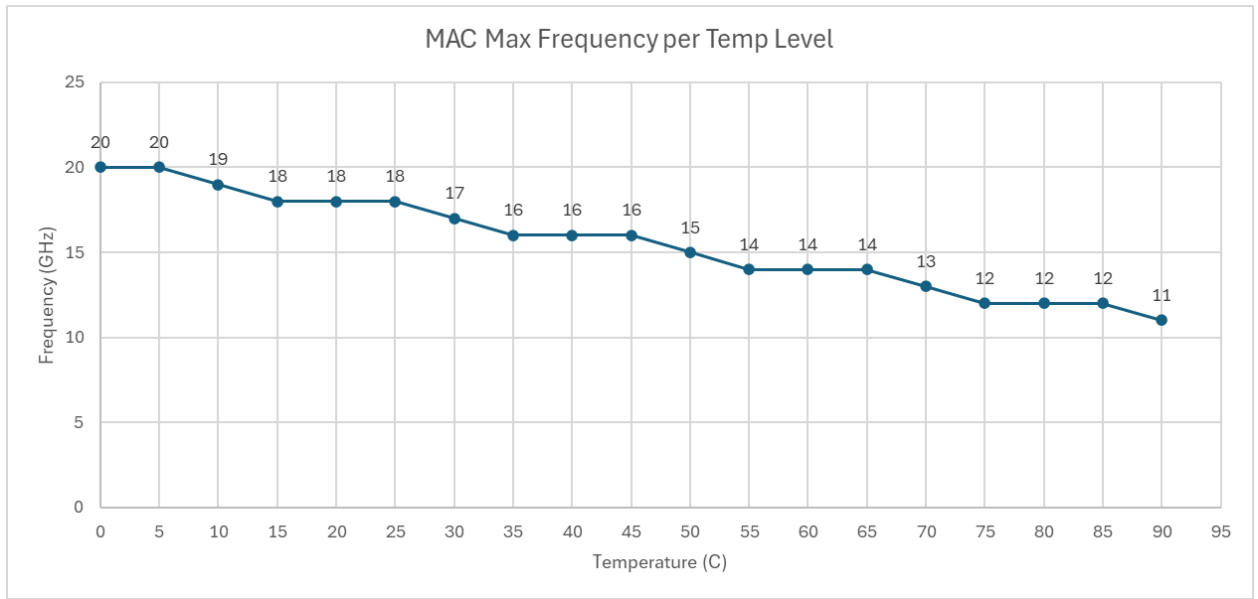


3. From the plot, the max frequency before failure is 18GHz at room temp. The delay is from the rising edge of the first clock cycle to the rising edge of output Y_0 with the input W=01, X=01.
4. The circuit component that fails to meet the timing constraints when increasing the frequency above 18 GHz is the 4-bit adder. If I were to redesign the circuit to be able to handle higher frequencies, I would replace the ripple-carry adder with a carry-lookahead adder to reduce the carry chain to $\log(N)$.

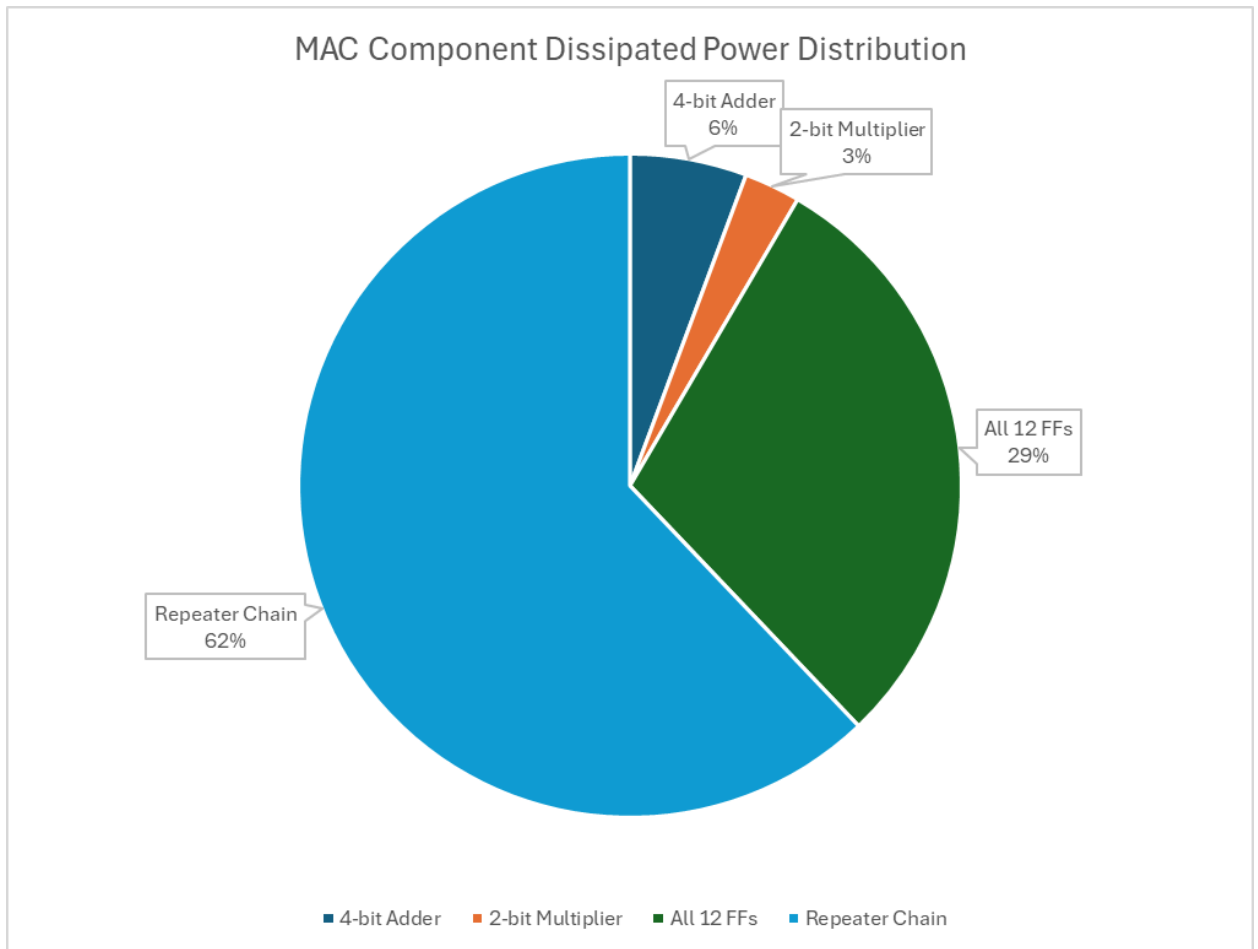


5. At a low VDD, the adder's critical path delay limits the MAC
At a high VDD, the register's hold time limits the MAC

6.



7.



Unit:	Pd (W)
4-bit Adder	1.42E-05
2-bit Multiplier	6.86E-06
All 12 FFs	4.91E-05
Repeater Chain	1.55E-04

If I were to redesign the entire system, I would minimize the total dissipated power by using fewer, larger stages in the repeater chain to sacrifice the delay for power dissipation. I would also replace the ripple-carry adder with a carry-save adder or a low-power carry-lookahead adder. I could lower the VDD as well.